

Name: _____ Date: _____ Period: _____

Inequalities: Testing Solutions

Solve each inequality. Select a value from the solution and test to find if the inequality is true.

Solve and Graph	Test Solution
$\frac{2g}{2} \geq \frac{16}{2}$ $g \geq 8$	To solve equation, divide by <u>2</u> Test value <u>9</u> $2(9) \geq 16$ $18 \geq 16$ <u>True</u> or False
$\frac{-2g}{-2} \geq \frac{16}{-2}$ $g \geq -8$	To solve equation, divide by <u>-2</u> Test value _____ $-2(-7) \geq 16$ $14 \geq 16$ True or <u>False</u>
$\frac{13}{1} \cdot \frac{m}{31} < 8 \cdot 3$ $m < 24$	To solve equation, multiply by <u>3</u> Test value <u>21</u> $\frac{(21)}{3} < 8$ $7 < 8$ <u>True</u> or False
$\frac{1-3}{1} \cdot \frac{m}{-31} < 8 \cdot -3$ $m < -24$	To solve equation, multiply by <u>-3</u> Test value <u>-27</u> $\frac{(-27)}{-3} < 8$ $9 < 8$ True or <u>False</u>

What do you notice about the test values when multiplying and dividing by a negative number? _____

The inequalities were false.

Inequality Rules for Multiplying and Dividing by a Negative Number



When multiplying or dividing both sides of an inequality by a **positive number**, the inequality remains true.



When multiplying or dividing both sides of an inequality by a **negative number**, the inequality will be false.
You must reverse or "flip" the symbol.

Solve each inequality.

$\frac{-12x}{-12} < \frac{36}{-12}$ Flip the sign? ☒ Yes ☐ No $x > -3$	$\frac{14n}{14} \geq \frac{-28}{14}$ Flip the sign? ☐ Yes ☒ No $n \geq -2$
$\frac{14}{1} \cdot \frac{f}{4} \geq -6 \cdot 4$ Flip the sign? ☐ Yes ☒ No $f \geq -24$	$\frac{1-3}{1} \cdot \frac{g}{-3} > 5 \cdot -3$ Flip the sign? ☒ Yes ☐ No $g < -15$
$\frac{-7k}{-7} \leq \frac{-35}{-7}$ Flip the sign? ☒ Yes ☐ No $k \geq 5$	$\frac{1-9}{1} \cdot \frac{m}{-9} < -2 \cdot -9$ Flip the sign? ☒ Yes ☐ No $m > 18$